MOTION CAPTURE PRODUCTION CHECKLIST

OPEN STANCE

Optical marker suit capture and animation coverage for Open Stance Gyeorugi.

VERSION	UPDATED	CHECKLIST ROWS	M1	M2	v1.0	FUTURE
0.5 draft	2026-10-29	288	29	44	189	26

Gameplay scope follows Docs/GDD_Open_Stance.docx v0.11 (2026-10-16). Optical marker suit capture is the production method selected on 2026-10-29; this supersedes the undecided capture-method text in GDD 16.1 and 18.1. M1 and M2 are early delivery priorities within v1.0; Future work never blocks them.

CONTENTS

HOW TO USE THIS LIST 3
SUMMARY BY CATEGORY 6
1 CALIBRATION AND REFERENCE 7
2 READY STANCES, IDLES AND GUARDS 8
3 FOOTWORK AND MOVEMENT 10
4 CHAMBERS 12
5 KICKS 14
6 PUNCHES AND FEINTS 18
7 DEFENSE AND CLINCH 19
8 HIT REACTIONS 21
9 FALLS, KNOCKDOWNS AND GET-UPS 22
10 TRANSITIONS AND RECOVERY 23
11 MATCH CEREMONY AND FLOW 25
12 RESULTS AND CELEBRATIONS 27
13 FATIGUE AND STAMINA 28
14 REFEREE AND OFFICIALS 29
15 COACH AND CORNER 31
16 MENUS AND CAREER (OUT OF GEAR) 32
17 TRAINING DRILLS 33
18 CROWDS AND FUTURE SPARRING FORMATS 35
19 SLIDING STRIKE VARIATIONS 36
CAPTURE SESSION LOG 38
CAPTURE REFERENCES 39

HOW TO USE THIS LIST

CHECKBOXES

COLUMN	TICK IT WHEN
Recorded	Every required variant has a usable take and a verified backup. For derived rows marked N/A, record the source clip IDs instead. Calibration rows use In engine = N/A.
Cleaned	Select and trim the take, correct marker labels and gaps before skeletal cleanup, then check contacts, pivots, spins and state endpoints. Derived rows require the assembled animation to pass the same checks.
In engine	Baked and retargeted to the matching skeleton; named correctly; tested in the relevant Montage, Blend Space or animation layer, including body, gear and transitions. Reference video alone is not completion.

PHASES

M1 Proof of concept (GDD 19 and 20.1): optical pipeline test, fighting idle, movement, visible stance switch, roundhouse, side/cut and spin side kick. Test spin hook tracking early; one captured roundhouse must replay in Unreal.

M2 Basic local match: lead/rear options, head kicks, punch, guards, reactions, clinch entry and separation, and match flow. If spins or jumps cannot be captured cleanly, use the GDD 16.4 and 20.1 placeholder fallback.

v1.0 Remaining capstone animation coverage: mapped techniques, all four guards, knockdowns, ceremony, officials, Career, grading, podium, menus and crowds. Jump variants normally reuse grounded captures plus shared launch and landing animation (GDD 16.4).

FUTURE After v1.0: unmapped techniques and clinch actions (GDD 9.4), video review, additional podium poses, team events, para and youth sets, and commentary. Do not schedule these before core coverage.

VARIANT SHORTHAND

TERM	MEANING
Left Lead & Right Lead	Capture Left Lead and Right Lead. Use the performer's left/right, not screen direction. Validate any approved mirroring for secondary motion in engine.
Lead leg & Rear leg	Front and back leg in each stance: four variant combinations before targets, movement and outcomes. A variant may need several takes; this is not a four-take session estimate.
Spin from Left Lead & Right Lead	Turning technique started from each stance; the spin direction follows the stance.
Front, Left, Right, Back	Incoming direction for a reaction. This does not make an illegal contact a scoring target; label legal and foul reactions separately.

TERM	MEANING
Loop	Capture several clean cycles. Trim and match pose, foot contacts and bounce phase in cleanup; raw first and last frames need not match exactly.
Two performers	Capture both roles with distinct calibrated subjects only after a visibility test. Otherwise capture roles separately with shared floor marks, distance, timing and reference video; align contacts in cleanup.

CAPTURE RULES FOR EVERY TAKE

- Use a fitted optical marker suit. The lab operator confirms the actual full-body marker template, anatomical placement and solver. Docs/Mocap_Marker_Set.md proposes 41 markers; it is not a confirmed lab template. Extra foot markers help only if the solve uses them.
- Before performers start, the lab operator calibrates the camera volume and floor and checks the working area for marker visibility. Confirm camera coverage and suitable capture settings with a slow and a full-speed kick, turn and jump; record the actual rate and settings rather than assuming a rate.
- Calibrate each subject at the start of the session using the lab template's required static pose and ROM. T- and A-poses below are references unless the lab specifies them for calibration. Repeat subject calibration if marker placement changes; keep the subject/template with its trials.
- Performers wear the marker suit for every category. Dobok, match gear, coach clothes and referee uniform describe the final character appearance. For gear handling, use a real or stand-in prop only after testing clearance and marker visibility; never cover required markers.
- Slate the session date, performer/subject, shot ID, full variant, take and calibration reference in the trial name/log and reference video. Sync reference cameras to optical trials with a visible clap or stomp, or verified timecode.
- Capture about one second of pre-roll and post-roll in the required source and destination states. Combat clips usually return to guard; chambers, holds, falls, get-ups and ceremony use their listed endpoints. Keep continuous bounce cycles for loop trimming.
- Capture 3-5 takes for each required variant as a starting plan; inspect a trial before repeating a block. Keep natural acceleration and impact timing. Log the selected take and pickups needed.
- Retarget across the five GDD body types, checking Compact (165 cm) and Tall (192 cm) first. Also test every shipping male/female target: the current library has nine presets. Check foot contacts, reach, joints, gear clearance and root motion.
- Capture stationary and sliding versions of mapped kicks and punches (GDD 7.4). Section 19 gives explicit sliding rows and all eight direction codes. Keep the listed stance, striking leg/hand and target; record the actual root and foot travel rather than translating a stationary clip over planted feet.
- Capture plausible bounce/chamber starts, rechambers and forward/back/switched landings for each technique. Do not require every landing or a static chamber for every spin/jump. Share transitions and log source clip IDs; do not duplicate the same recording for CHM, KCK and TRN rows.
- Jump side, roundhouse, hook, axe and spin hook are derived from their grounded technique plus shared launch/landing (TRN-12). Keep their rows for assembly and engine acceptance; bespoke jump captures are optional pickups if reuse fails. The Tornado Kick requires its own capture.

- Record front and side reference video where practical; these cameras are references, not the optical solve. Log frame rate and sync offset. Record contact frames, planted feet, lift-off/landing and turns for cleanup.
- Inspect marker reconstruction, swaps and gaps before leaving the lab, especially during crossed limbs, spins, high kicks, floor work, props and partner contact. Fix labels before filling gaps; re-record obscured impact or landing when a credible solve cannot be recovered.
- Evaluate supporting-foot pivot, hip rotation, chamber, extension, rechamber, balance, landing and root trajectory. Do not smooth away kick acceleration or lock a foot that should pivot.
- The proposed body marker set does not capture finger articulation, toe curl or facial expressions. Hand-key these or use separate capture. This applies to fists, belt tying, glove/helmet handling, referee finger counts, clapping, speech and expressions.
- Falls use crash mats and performers able to fall safely; use controlled pad contact or acted reactions. Do not strike heads or groins to obtain reactions. Check mat/prop height and marker clearance before floor or partner work.
- Before leaving the lab, back up raw trials, calibrated subjects/templates, labeled processed calibration and motion C3D, reference videos, placement diagram, measurements and session settings. Collect solved skeleton FBX if available. Log units, axes, FPS and frame range.
- Prove one trial end to end: inspect the source solve, clean and bake in MotionBuilder, import to the matching MetaHuman skeleton or retarget a separate source, then save/reopen in Unreal. Check bare body, dobok and gear without a live capture connection. Marker-only C3D is not a finished animation.

NAMING

Raw trials: OS_<SessionDate>_<Subject>_<ID>_<ShortName>_<Variant>_T<take>, e.g. OS_20261029_P01_KCK-01_RoundBody_RearLeg_LeftLead_Stationary_Pad_T03. Include target, movement, outcome and landing in the variant/log when relevant. Unreal assets use underscores instead of hyphens (KCK_01) and omit the take suffix; keep a manifest linking each asset to its selected source trial and frame range. Sliding variants add SlideF, SlideB, SlideL, SlideR, SlideFL, SlideFR, SlideBL or SlideBR; use Still for the stationary baseline. For example, SLD-03_RoundBody_RearLeg_LeftLead_SlideF_Pad. Record slide distance and duration in the take manifest.

Counts are checklist rows, not recordings, takes or unique exported animations. Each row expands into its stated variants; calibration/reference and derived/reused rows are included. Shared source clips may satisfy several rows. Schedule only the needed phases and approved variants. SLD rows track sliding coverage separately from their stationary KCK/PNF sources; existing skip and counter captures can satisfy a direction when the motion matches.

SUMMARY BY CATEGORY

#	CATEGORY	ID PREFIX	ROWS	M1	M2	v1.0	FUTURE
1	Calibration and reference	CAL	10	10	-	-	-
2	Ready stances, idles and guards	IDL	19	2	7	9	1
3	Footwork and movement	FTW	33	4	6	23	-
4	Chambers	CHM	15	6	2	7	-
5	Kicks	KCK	34	5	2	15	12
6	Punches and feints	PNF	11	-	1	7	3
7	Defense and clinch	DEF	16	-	8	6	2
8	Hit reactions	HIT	14	-	4	10	-
9	Falls, knockdowns and get-ups	FAL	14	-	-	14	-
10	Transitions and recovery	TRN	13	-	6	7	-
11	Match ceremony and flow	CER	18	-	4	13	1
12	Results and celebrations	RES	13	-	-	12	1
13	Fatigue and stamina	FAT	5	-	-	5	-
14	Referee and officials	REF	16	-	-	15	1
15	Coach and corner	COA	7	-	-	7	-
16	Menus and career (out of gear)	MNU	15	-	-	15	-
17	Training drills	TRG	15	-	-	15	-
18	Crowds and future sparring formats	FUT	6	-	-	1	5
19	Sliding strike variations	SLD	14	2	4	8	-
	Total		**288**	**29**	**44**	**189**	**26**

1 CALIBRATION AND REFERENCE

Per performer/session: record the lab-required calibration and references before action trials. Repeat subject calibration after marker changes. The lab operator handles camera/floor calibration separately. These rows are references, not game animation assets.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	N/A	CAL-01	Static calibration and T-pose reference	Lab-required static pose; T-pose reference 5 s	M1	Use the actual template pose; T-pose alone is not universal subject calibration.
☐	☐	N/A	CAL-02	A-pose reference	5 s hold if useful for target fitting	M1	Reference only; fit to the actual target rest pose and axes, not an assumed MetaHuman pose.
☐	☐	N/A	CAL-03	Sync clap and stomp	Start and end of each take block	M1	Sync optical trial and reference video; log the frame/time offset.
☐	☐	N/A	CAL-04	Range of motion	Neck, shoulders, elbows, wrists, spine, hips, knees, ankles	M1	Lab-directed ROM; check labels and solve across the full movement range.
☐	☐	N/A	CAL-05	Neutral relaxed stand	10 s	M1	Base pose for comparison
☐	☐	N/A	CAL-06	Walk and jog reference	Forward, backward and across volume	M1	Check scale, floor, foot contacts and root travel.
☐	☐	N/A	CAL-07	High kick flexibility reference	Slow front, side and round kicks to max height	M1	Controlled comfortable range; inspect high-kick marker visibility and solve.
☐	☐	N/A	CAL-08	Spin tracking test	Slow 360° turn, then fast spin	M1	Tests occlusion handling before real spinning kicks
☐	☐	N/A	CAL-09	Performer measurements and placement reference	Measurements required by the lab; front/back placement photos	M1	Log subject ID, marker template, offsets, capture rate, units and axes.
☐	☐	N/A	CAL-10	Optical visibility and export test	Guard, step, chamber, kick, fast turn and controlled jump	M1	Inspect labels/gaps and export one test through MotionBuilder to Unreal before bulk capture.

2 READY STANCES, IDLES AND GUARDS

Capture at least 10 seconds of clean cycles per idle. Trim loops to matching contacts and bounce phase. High, Cross, Mixed and Low guard postures remain undefined in GDD 8.7: approve reference poses before final capture. The starting guard reuses one of those four, not a fifth guard.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	IDL-01	Sparring guard bounce (starting guard)	Left Lead & Right Lead	M1	Primary idle, in the fighter's starting guard (GDD 10.1); loop
☐	☐	☐	IDL-02	Neutral guard bounce, tired	Left Lead & Right Lead	M2	Lower hands, slower bounce; loop
☐	☐	☐	IDL-03	Neutral guard bounce, exhausted	Left Lead & Right Lead	v1.0	Heavy breathing, flat-footed; loop
☐	☐	☐	IDL-04	High guard bounce	Left Lead & Right Lead	M2	Posture still to be defined (GDD 8.7); loop
☐	☐	☐	IDL-05	Cross guard bounce	Left Lead & Right Lead	M2	Posture still to be defined (GDD 8.7); loop
☐	☐	☐	IDL-06	Mixed guard bounce	Left Lead & Right Lead	v1.0	Posture still to be defined (GDD 8.7); loop
☐	☐	☐	IDL-07	Flat-footed guard (no bounce)	Left Lead & Right Lead	v1.0	Used after stoppages and at low stamina; loop
☐	☐	☐	IDL-08	Guard change: starting guard ↔ High	Both directions; Left Lead & Right Lead	M2	
☐	☐	☐	IDL-09	Guard change: starting guard ↔ Cross	Both directions; Left Lead & Right Lead	M2	
☐	☐	☐	IDL-10	Guard change: High ↔ Cross	Both directions; Left Lead & Right Lead	v1.0	
☐	☐	☐	IDL-11	Guard change: to and from Mixed and Low	Mixed/Low to and from each other guard; both stances	v1.0	Include Mixed <-> Low and both directions; deduplicate the starting-guard pair.
☐	☐	☐	IDL-12	Stance switch in place (bounce switch)	Left→Right and Right→Left	M1	A button; visible switch, required by M1 stance/leg story (GDD 8.13).
☐	☐	☐	IDL-13	Stance switch while advancing	Left→Right and Right→Left	v1.0	

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	IDL-14	Stance switch while retreating	Left→Right and Right→Left	v1.0	
☐	☐	☐	IDL-15	Hop switch (double switch feint)	Left→Right→Left and Right→Left→Right	FUTURE	
☐	☐	☐	IDL-16	Joon-bi ready stance hold	Hands in ready position; 5 s	M2	Before shi-jak
☐	☐	☐	IDL-17	Charyeot attention stance hold	5 s	M2	
☐	☐	☐	IDL-18	Relaxed standing idle (no guard)	Weight shifts; 20 s	v1.0	During stoppages and menus
☐	☐	☐	IDL-19	Low guard bounce	Left Lead & Right Lead	v1.0	Normal upright guard choice, not a crouching foul; posture still needs approval (GDD 8.7).

3 FOOTWORK AND MOVEMENT

Record each move as a single step and as a repeated loop, in guard. These feed the movement Blend Space, so keep speed and bounce consistent. FTW-15/16 cover forward/back slides; FTW-32/33 add lateral/diagonal slides. These movement clips support, but do not replace, the sliding strike captures in section 19.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	FTW-01	Step forward, small	Left Lead & Right Lead	M1	Single step; returns to bounce
☐	☐	☐	FTW-02	Step back, small	Left Lead & Right Lead	M1	
☐	☐	☐	FTW-03	Advance loop (repeated steps forward)	Left Lead & Right Lead; normal and fast	M1	Blend Space loop
☐	☐	☐	FTW-04	Retreat loop (repeated steps back)	Left Lead & Right Lead; normal and fast	M1	Blend Space loop
☐	☐	☐	FTW-05	Step forward, long	Left Lead & Right Lead	M2	Closes distance for body kicks
☐	☐	☐	FTW-06	Step back, long	Left Lead & Right Lead	M2	Out of range
☐	☐	☐	FTW-07	Side step left	Left Lead & Right Lead	M2	
☐	☐	☐	FTW-08	Side step right	Left Lead & Right Lead	M2	
☐	☐	☐	FTW-09	Strafe loop left	Left Lead & Right Lead	v1.0	Blend Space loop
☐	☐	☐	FTW-10	Strafe loop right	Left Lead & Right Lead	v1.0	Blend Space loop
☐	☐	☐	FTW-11	Circle left (pivot around opponent)	Left Lead & Right Lead	v1.0	Loop
☐	☐	☐	FTW-12	Circle right (pivot around opponent)	Left Lead & Right Lead	v1.0	Loop
☐	☐	☐	FTW-13	Diagonal step forward-left and forward-right	Left Lead & Right Lead	v1.0	Angle entries
☐	☐	☐	FTW-14	Diagonal step back-left and back-right	Left Lead & Right Lead	v1.0	
☐	☐	☐	FTW-15	Slide step in (drag rear foot)	Forward; both stances; single slide and loop	v1.0	Taekwondo slide, not a walk
☐	☐	☐	FTW-16	Slide step out	Backward; both stances; single slide and loop	v1.0	

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	FTW-17	Dash forward (quick step)	Left Lead & Right Lead	M2	B button; explosive; closing to contact range triggers a clinch automatically into DEF-12 (GDD 6.4)
☐	☐	☐	FTW-18	Dash back (quick step)	Left Lead & Right Lead	M2	
☐	☐	☐	FTW-19	Dash side left and right	Left Lead & Right Lead	v1.0	
☐	☐	☐	FTW-20	Pivot turn 45° left and right	Left Lead & Right Lead	v1.0	Re-angle to opponent
☐	☐	☐	FTW-21	Pivot turn 90° left and right	Left Lead & Right Lead	v1.0	
☐	☐	☐	FTW-22	Turn around 180° to face opponent	Clockwise and counter-clockwise	v1.0	After being circled or after a missed spin
☐	☐	☐	FTW-23	Hop in place	Left Lead & Right Lead	v1.0	
☐	☐	☐	FTW-24	Stop from advance	Left Lead & Right Lead	v1.0	Transition to bounce
☐	☐	☐	FTW-25	Stop from retreat	Left Lead & Right Lead	v1.0	Transition to bounce
☐	☐	☐	FTW-26	Start advance from bounce	Left Lead & Right Lead	v1.0	
☐	☐	☐	FTW-27	Start retreat from bounce	Left Lead & Right Lead	v1.0	
☐	☐	☐	FTW-28	Boundary check (retreat stops at line, looks down)	Left Lead & Right Lead	v1.0	Near the out-of-bounds line
☐	☐	☐	FTW-29	Walk forward (no guard)	Loop; start and stop	v1.0	Entering and leaving the mat, menus
☐	☐	☐	FTW-30	Walk turn left and right (no guard)	90° and 180°	v1.0	
☐	☐	☐	FTW-31	Jog in place and warm-up bounce (no guard)	Loop	v1.0	Pre-match warm-up
☐	☐	☐	FTW-32	Slide left and right	Both directions; Left Lead & Right Lead; single and loop	v1.0	Maintain guard/facing. Capture actual travel and contacts; distinguish slide from dash or stance switch.
☐	☐	☐	FTW-33	Slide diagonally	Forward-left/right and back-left/right; both stances	v1.0	Single and repeated travel; log displacement and duration. Keep natural footwork and consistent facing.

4 CHAMBERS

The chamber is the core of the Skill Stick system. Every chamber needs enter, hold loop and exit so techniques can branch from it.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	CHM-01	Lead-leg chamber enter	Left Lead & Right Lead	M1	From bounce to knee up
☐	☐	☐	CHM-02	Lead-leg chamber hold loop	Left Lead & Right Lead	M1	Balanced; small hops allowed; loop
☐	☐	☐	CHM-03	Lead-leg chamber cancel (return to guard)	Left Lead & Right Lead	M1	Fake attack: hold LB, then release (GDD 8.6)
☐	☐	☐	CHM-04	Rear-leg chamber enter	Left Lead & Right Lead	M1	
☐	☐	☐	CHM-05	Rear-leg chamber hold loop	Left Lead & Right Lead	M1	Loop
☐	☐	☐	CHM-06	Rear-leg chamber cancel	Left Lead & Right Lead	M1	
☐	☐	☐	CHM-07	High chamber enter and hold (head-kick height)	Lead leg & Rear leg; Left Lead & Right Lead	M2	Knee higher; loop
☐	☐	☐	CHM-08	Chamber hop forward	Lead leg & Rear leg; Left Lead & Right Lead	v1.0	Skipping while chambered
☐	☐	☐	CHM-09	Chamber hop back	Lead leg & Rear leg; Left Lead & Right Lead	v1.0	
☐	☐	☐	CHM-10	Chamber drift (turn in place while chambered)	Lead leg & Rear leg; Left Lead & Right Lead	v1.0	Re-aims the knee
☐	☐	☐	CHM-11	Chamber to feint (fake extension, rechamber)	Lead leg & Rear leg; Left Lead & Right Lead	v1.0	
☐	☐	☐	CHM-12	Chamber to stance switch landing	Lead leg & Rear leg; Left Lead & Right Lead	v1.0	Lands in the other stance
☐	☐	☐	CHM-13	Rechamber after kick (retract to knee)	Every mapped kick; its scheduled phase	M2	Extract from kick takes when available; do not wait until M2 for M1 kick recovery.
☐	☐	☐	CHM-14	Spin chamber (back-turn wind-up) enter and cancel	Spin from Left Lead & Right Lead	v1.0	Start of spinning kicks

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	CHM-15	Slide while chambered	Eight directions; lead/rear chamber; both stances	v1.0	Enter chamber, travel, hold or cancel. Use short natural shuffles/hops where needed; preserve support-foot contacts. Compare CHM-08/09.

5 KICKS

Rows cover technique families and their variants. Use the capture rules above to plan feasible targets, movement and outcomes; retain one trial as the source for shared transitions. Jump rows marked N/A in Recorded are derived by default. Body/head values are game-design values from GDD 8.3 and 9; animation alone does not establish a legal score. Use section 19 for sliding kick coverage; an entry step followed by a stationary strike does not establish a slide during the strike.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	KCK-01	Roundhouse kick (dollyo chagi), body	Lead leg & Rear leg; Left Lead & Right Lead	M1	Core scoring kick; 2 points. Lead leg by default, rear leg with RT (GDD 9.1)
☐	☐	☐	KCK-02	Round kick, body, fast lead leg (bandal chagi)	Lead leg; Left Lead & Right Lead	FUTURE	Fast lead-leg roundhouse: not yet mapped (GDD 9.4)
☐	☐	☐	KCK-03	Round kick, head	Rear leg; Left Lead & Right Lead	M2	3 points
☐	☐	☐	KCK-04	Round kick, head, lead leg	Lead leg; Left Lead & Right Lead	v1.0	
☐	☐	☐	KCK-05	Skip round kick (slide in, lead leg)	Lead leg; Left Lead & Right Lead	v1.0	Closes long distance
☐	☐	☐	KCK-06	Counter round kick while stepping back	Lead leg & Rear leg; Left Lead & Right Lead	v1.0	Back-step counter
☐	☐	☐	KCK-07	Round kick counter to a turning opponent	Rear leg; both stances	v1.0	Aim at a legal trunk/side target; contact on the back is not automatically a score.
☐	☐	☐	KCK-08	Cut kick (lead-leg side-on jam), body	Lead leg; Left Lead & Right Lead	M1	Stops advancing opponents
☐	☐	☐	KCK-09	Side kick (yeop chagi), body and head	Lead & rear leg; both stances; body and head	M1	M1 body priority; add head coverage for M2/v1.0. Body 2, head 3; RT rear leg, RB head (GDD 9.1).
☐	☐	☐	KCK-10	Push kick (mireo chagi), body	Lead leg & Rear leg; Left Lead & Right Lead	FUTURE	Not yet mapped (GDD 9.4)
☐	☐	☐	KCK-11	Front kick (ap chagi), body	Lead leg & Rear leg; Left Lead & Right Lead	FUTURE	Not yet mapped (GDD 9.4)

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	KCK-12	Axe kick (naeryo chagi), head	Lead leg & Rear leg; Left Lead & Right Lead	v1.0	Straight up and down
☐	☐	☐	KCK-13	Hook kick (huryeo chagi), body and head	Lead & rear leg; both stances; body and head	v1.0	Body 2, head 3. RS opposite the kicking leg; RT changes leg and mirrors direction (GDD 9.1).
☐	☐	☐	KCK-14	Spin side kick (back kick, dwi chagi)	Spin from Left Lead & Right Lead; body and head	M1	Turning kick; 4 points body, 6 head (GDD 9.2)
☐	☐	☐	KCK-15	Spin side kick, counter (opponent advancing)	Spin from Left Lead & Right Lead	M2	Most common spinning counter
☐	☐	☐	KCK-16	Spin hook kick (dwi huryeo chagi)	Spin from Left Lead & Right Lead; head and body	M1	M1 feasibility test with roundhouse (GDD 19); expand targets/variants for M2. Body 4, head 6 when registered as spinning.
☐	☐	☐	KCK-17	360° back round kick (dwi dollyo chagi), body	Spin from Left Lead & Right Lead	FUTURE	
☐	☐	☐	KCK-18	360° back round kick, head	Spin from Left Lead & Right Lead	FUTURE	
☐	☐	☐	KCK-19	Double kick (dubal dangseong), body–body	Starting lead and rear; both stances	FUTURE	Not yet mapped (GDD 9.4)
☐	☐	☐	KCK-20	Double kick, body–head	Starting lead and rear; both stances	FUTURE	
☐	☐	☐	KCK-21	Triple round kick flurry	Both stances	FUTURE	
☐	☐	☐	KCK-22	Switch round kick (switch then rear-leg round kick)	Both stances	v1.0	
☐	☐	☐	KCK-23	Tornado kick (dolgae chagi), body and head	From Left Lead & Right Lead; body and head	v1.0	Unique jumping turning kick capture. Y then RS half circle down + half circle up; RB head. Input circle is not a prescribed 540-degree body rotation. Body 4, head 6 (GDD 9.2).
☐	☐	☐	KCK-24	Jumping back kick (twio dwi chagi), body	Spin from Left Lead & Right Lead	FUTURE	

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
N/A	☐	☐	KCK-25	Jump spin hook kick (dwi huryeo chagi), head and body	Spin from Left Lead & Right Lead	v1.0	DERIVED: KCK-16 + TRN-12 shared launch/landing (GDD 16.4). Optional capture if reuse fails. Y, then Hold RT + RS Half Circle Down; RB for the head (GDD 7.2, 9.2)
N/A	☐	☐	KCK-26	Jump round kick, body and head	Lead leg & Rear leg; Left Lead & Right Lead	v1.0	DERIVED: KCK-01/03/04 + TRN-12 shared launch/landing (GDD 16.4). Optional capture if reuse fails. Y, then RS Left/Right, toward the lead leg; RT mirrors it to the back leg; RB for the head (GDD 7.2, 9.2)
☐	☐	☐	KCK-27	In-close round kick (fan kick / pull-back), head	Lead leg & Rear leg; Left Lead & Right Lead	FUTURE	Close-range clinch kick: not yet mapped (GDD 9.4)
☐	☐	☐	KCK-28	In-close axe kick, head	Lead leg & Rear leg; Left Lead & Right Lead	FUTURE	Close-range clinch kick: not yet mapped (GDD 9.4)
☐	☐	☐	KCK-29	Low hook kick in clinch (tagging the back of the head)	Lead leg & Rear leg; Left Lead & Right Lead	FUTURE	
☐	☐	☐	KCK-30	Kick miss overshoot recovery	For round kick and spinning kicks	v1.0	Overbalanced after a miss; recover to guard
☐	☐	☐	KCK-31	Spinning kick miss, lose orientation	Spin from Left Lead & Right Lead	v1.0	Turns too far; resets facing
N/A	☐	☐	KCK-32	Jump side kick (yeop chagi), body and head	Lead leg & Rear leg; Left Lead & Right Lead	v1.0	DERIVED: KCK-09 + TRN-12 shared launch/landing (GDD 16.4). Optional capture if reuse fails. Y, then RS Up; RT for the back leg; RB for the head (GDD 7.2, 9.2)

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
N/A	☐	☐	KCK-33	Jump hook kick (huryeo chagi), head and body	Lead leg & Rear leg; Left Lead & Right Lead	v1.0	DERIVED: KCK-13 + TRN-12 shared launch/landing (GDD 16.4). Optional capture if reuse fails. Y, then RS Left/Right, toward the side opposite the lead leg; RT mirrors it to the lead leg's side; RB for the head (GDD 7.2, 9.2)
N/A	☐	☐	KCK-34	Jump axe kick (naeryo chagi), head	Lead leg & Rear leg; Left Lead & Right Lead	v1.0	DERIVED: KCK-12 + TRN-12 shared launch/landing (GDD 16.4). Optional capture if reuse fails. Y, then RS Left/Right Quarter Circle Forward, Down (GDD 7.2, 9.2)

6 PUNCHES AND FEINTS

Punches score only to the trunk protector. Feints must look like the start of a real technique.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	PNF-01	Straight punch (jireugi), rear hand, body	Left Lead & Right Lead	M2	X; trunk only, 1 point. Default straight punch for the basic match; pose fingers separately.
☐	☐	☐	PNF-02	Straight punch, lead hand, body	Left Lead & Right Lead	v1.0	
☐	☐	☐	PNF-03	Punch while stepping back	Left Lead & Right Lead	v1.0	Creates space
☐	☐	☐	PNF-04	Repeated straight punches, body	Left Lead & Right Lead	v1.0	Reference for repeated X presses; reuse single punches and recovery. No new double-punch input or combo string.
☐	☐	☐	PNF-05	Punch to break a clinch (push-punch)	Left Lead & Right Lead	FUTURE	
☐	☐	☐	PNF-06	Lead-leg lift feint	Left Lead & Right Lead	v1.0	Generic LB fake may reuse chamber enter/cancel. Check startup/cancel for every mapped technique, including jump and spin fakes (GDD 7.10).
☐	☐	☐	PNF-07	Rear-leg lift feint	Left Lead & Right Lead	v1.0	
☐	☐	☐	PNF-08	Step-in feint (half step, stop)	Left Lead & Right Lead	v1.0	
☐	☐	☐	PNF-09	Shoulder / hip turn feint (fake spin)	Left Lead & Right Lead	v1.0	Sells a spin side kick
☐	☐	☐	PNF-10	Switch feint (fake switch, stay)	Left Lead & Right Lead	FUTURE	
☐	☐	☐	PNF-11	Head / upper-body fake	Left Lead & Right Lead	FUTURE	

7 DEFENSE AND CLINCH

Use controlled pad contact or acted block recoil. Test partner visibility before paired work; capture roles separately if needed. Include enter, brief hold, impact response and recovery. LT body cover, LT + RS Up high cover, LT + RS Down leg check and LT + RS Left/Right evade all need coverage (GDD 8.8).

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	DEF-01	Arm block, body, lead side	Left Lead & Right Lead	M2	Hold LT alone closes hands/arms against punches and body kicks; not a dropped/open guard (GDD 8.8).
☐	☐	☐	DEF-02	Arm block, body, rear side	Left Lead & Right Lead	M2	
☐	☐	☐	DEF-03	Arm block, head, lead side	Left Lead & Right Lead	M2	Forearm raises to cover head; Active Defense high block (Hold LT + RS Up, GDD 8.8)
☐	☐	☐	DEF-04	Arm block, head, rear side	Left Lead & Right Lead	M2	
☐	☐	☐	DEF-05	Double-arm cover (both arms protect head)	Left Lead & Right Lead	v1.0	
☐	☐	☐	DEF-06	Parry a round kick (redirect with hand)	Left Lead & Right Lead	FUTURE	
☐	☐	☐	DEF-07	Leg check / lift leg to block	Lead leg; Left Lead & Right Lead	v1.0	Active Defense leg check (Hold LT + RS Down, GDD 8.8); short only — held too long it's the leg-lifting penalty (GDD 8.4)
☐	☐	☐	DEF-08	Evade: step back out of range	Left Lead & Right Lead	M2	Distance defense
☐	☐	☐	DEF-09	Evade: lean back (sway)	Sway left/right and off-line; both stances	v1.0	LT + RS Left/Right; include lateral sway or step off line, not only a backward lean (GDD 8.8).
☐	☐	☐	DEF-10	Evade: side step and pivot	Left and right; both stances	v1.0	Sets up counters

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	DEF-11	Evade: duck / crouch	Left Lead & Right Lead	v1.0	Penalty behaviour reference only; no mapped duck attack/defense input. Do not treat normal Low Guard as a foul.
☐	☐	☐	DEF-12	Clinch enter (bodies meet, arms tangle)	Left Lead & Right Lead	M2	Two roles; automatic contact state from dash or both advancing (GDD 6.4), not an extra clinch command.
☐	☐	☐	DEF-13	Clinch hold loop (pushing for position)	Left Lead & Right Lead	M2	Two roles; contact idle only. Do not add unmapped grabbing or clinch attacks (GDD 9.4).
☐	☐	☐	DEF-14	Clinch break (kal-yeo separation)	Left Lead & Right Lead	M2	Two roles; kal-yeo step-back separation. Use shared marks and timing if recorded separately.
☐	☐	☐	DEF-15	Clinch pivot (turn opponent in clinch)	Left and right	FUTURE	Clinch action: not yet mapped (GDD 9.4)
☐	☐	☐	DEF-16	Turn away from attack (back to opponent)	Left Lead & Right Lead	v1.0	Penalty behaviour

8 HIT REACTIONS

Performers react to pad contact or a partner's controlled touch; exaggerate slightly. Record every reaction from both stances and return to guard.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	HIT-01	Minor reaction, trunk	Direction: Front, Left, Right, Back	M2	Additive-ready; short
☐	☐	☐	HIT-02	Minor reaction, head	Direction: Front, Left, Right, Back	M2	Acted/reference response; no head impact required. Additive conversion in cleanup.
☐	☐	☐	HIT-03	Normal reaction, trunk	Direction: Front, Left, Right, Back	M2	Partial body; 1 step
☐	☐	☐	HIT-04	Normal reaction, head	Direction: Front, Left, Right, Back	M2	
☐	☐	☐	HIT-05	Heavy reaction, trunk (stagger)	Direction: Front, Left, Right, Back	v1.0	2–3 steps; recover to guard
☐	☐	☐	HIT-06	Heavy reaction, head (stagger)	Direction: Front, Left, Right, Back	v1.0	
☐	☐	☐	HIT-07	Loss of balance and recover	Forward, back, left, right	v1.0	Arms out, regains stance
☐	☐	☐	HIT-08	Hit while chambered (knocked off chamber)	Lead leg & Rear leg; Left Lead & Right Lead	v1.0	
☐	☐	☐	HIT-09	Hit during spin (interrupted spinning kick)	Spin from Left Lead & Right Lead	v1.0	
☐	☐	☐	HIT-10	Kick blocked on arm (attacker's reaction)	Per kick family: round, side, spinning	v1.0	Recoil into guard
☐	☐	☐	HIT-11	Block absorbs heavy kick (defender pushed back)	Left Lead & Right Lead	v1.0	
☐	☐	☐	HIT-12	Punched in trunk (minor push back)	Left Lead & Right Lead	v1.0	
☐	☐	☐	HIT-13	Groin / illegal low kick reaction	Left Lead & Right Lead	v1.0	Acted foul response; no actual groin contact.
☐	☐	☐	HIT-14	Standing count reaction (dazed, recover)	Left Lead & Right Lead	v1.0	Referee count after a heavy head kick

9 FALLS, KNOCKDOWNS AND GET-UPS

Capture on crash mats with performers who can fall safely. No performer should be struck hard.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	FAL-01	Knockdown fall backward	From head kick and trunk kick	v1.0	Captured on mats
☐	☐	☐	FAL-02	Knockdown fall forward		v1.0	
☐	☐	☐	FAL-03	Knockdown fall to the side	Left and right	v1.0	
☐	☐	☐	FAL-04	Spinning / off-balance fall	Left and right	v1.0	
☐	☐	☐	FAL-05	Slip / trip on mat (no contact)	Left Lead & Right Lead	v1.0	Non-hit fall path; game logic classifies Falling penalty separately from hit-triggered knockdown (GDD 8.10).
☐	☐	☐	FAL-06	Fall after a missed spinning kick	Spin from Left Lead & Right Lead	v1.0	
☐	☐	☐	FAL-07	Tangled-legs fall in clinch	Two performers	v1.0	
☐	☐	☐	FAL-08	Grounded idle on back	Loop	v1.0	
☐	☐	☐	FAL-09	Grounded idle on front	Loop	v1.0	
☐	☐	☐	FAL-10	Grounded idle on side	Left and right; loop	v1.0	
☐	☐	☐	FAL-11	Get-up from back	Quick and slow (hurt)	v1.0	Ends in guard
☐	☐	☐	FAL-12	Get-up from front	Quick and slow	v1.0	
☐	☐	☐	FAL-13	Get-up from side	Left and right	v1.0	
☐	☐	☐	FAL-14	Sit up and check injury, then stand		v1.0	Stoppage flow

10 TRANSITIONS AND RECOVERY

Coverage rows for linking states. Extract usable pieces from longer takes and log the source frame range; capture pickups only where the source is missing or fails to blend. Never count a shared clip as a new capture requirement.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	TRN-01	Kick landing forward to guard	Per kick family; both stances	M2	Kicking foot lands ahead into the logged destination stance; not always the original lead foot.
☐	☐	☐	TRN-02	Kick landing back to guard (return foot)	Per kick family; both stances	M2	
☐	☐	☐	TRN-03	Kick landing in switched stance	Per kick family	v1.0	
☐	☐	☐	TRN-04	Rechamber into second kick	Round→round, round→head, cut→round	v1.0	Player-triggered follow-up, not a fixed combo string; share rechamber clips where possible.
☐	☐	☐	TRN-05	Spin landing to guard, facing opponent	Spin from Left Lead & Right Lead	M2	
☐	☐	☐	TRN-06	Recover from dash to bounce	Left Lead & Right Lead	v1.0	
☐	☐	☐	TRN-07	Recover from clinch break to bounce	Left Lead & Right Lead	v1.0	
☐	☐	☐	TRN-08	Guard to relaxed stand (on kal-yeo / keu-man)	Left Lead & Right Lead	M2	Referee stops action
☐	☐	☐	TRN-09	Relaxed stand to guard (on kye-sok)	Left Lead & Right Lead	M2	Action resumes
☐	☐	☐	TRN-10	Return to start mark (walk back, turn, face)		v1.0	After a stoppage
☐	☐	☐	TRN-11	Joon-bi to fighting guard (on shi-jak)	Left Lead & Right Lead	M2	Match start

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	TRN-12	Shared jump launch and landing	Both stances; grounded takeoff, airborne transition, landing to guard	v1.0	Layer/blend under grounded side, roundhouse, hook, axe and spin hook. Preserve lift-off/landing timing, root travel and foot contacts; bespoke capture only if needed (GDD 16.4).
☐	☐	☐	TRN-13	Sliding attack recovery and cancel	Eight directions; both stances; hit, miss, block and fake cancel	v1.0	Per technique family: continue travel or stop into guard. Reuse SLD take tails; capture pickups only for missing momentum/foot transitions.

11 MATCH CEREMONY AND FLOW

Final characters wear match gear. Performers stay in optical marker suits. Mimic gear clearance and use visibility-tested props for adjustment actions; do not obscure body markers with a dobok or protector.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	CER-01	Walk onto the mat to start mark	Chung side and Hong side	M2	
☐	☐	☐	CER-02	Charyeot (attention) on command		M2	
☐	☐	☐	CER-03	Kyeongrye (bow) to opponent		M2	
☐	☐	☐	CER-04	Kyeongrye to referee		v1.0	
☐	☐	☐	CER-05	Joon-bi (ready) on command		M2	
☐	☐	☐	CER-06	Walk to corner at round end		v1.0	
☐	☐	☐	CER-07	Sit in corner chair		v1.0	
☐	☐	☐	CER-08	Seated rest between rounds (loop)	Fresh and tired	v1.0	Breathing, drinking water
☐	☐	☐	CER-09	Listen to coach (nodding, seated)		v1.0	
☐	☐	☐	CER-10	Stand up from chair and walk back to mark		v1.0	
☐	☐	☐	CER-11	Adjust helmet		v1.0	
☐	☐	☐	CER-12	Adjust chest protector straps		v1.0	
☐	☐	☐	CER-13	Adjust gloves / foot guards		v1.0	
☐	☐	☐	CER-14	Tie / fix belt		v1.0	Share source with MNU-07. Fingers and belt contacts need hand animation/IK beyond the body solve.
☐	☐	☐	CER-15	Handshake / glove touch with opponent	Two performers	v1.0	Start and end of match
☐	☐	☐	CER-16	Bow to coach after match		v1.0	
☐	☐	☐	CER-17	Walk off the mat	Winner and loser versions	v1.0	

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	CER-18	Coach video review request (raise card)	Coach performer	FUTURE	Video review (GDD 8.11)

12 RESULTS AND CELEBRATIONS

Respectful celebrations only; no taunts.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	RES-01	Scoring reaction (small fist pump, back to guard)	Left Lead & Right Lead	v1.0	After scoring
☐	☐	☐	RES-02	Frustration after conceding (short, controlled)	Left Lead & Right Lead	v1.0	
☐	☐	☐	RES-03	Round won (nod, step back)		v1.0	
☐	☐	☐	RES-04	Round lost (exhale, reset)		v1.0	
☐	☐	☐	RES-05	Winner's hand raised by referee	Fighter/referee roles; Chung and Hong winners	v1.0	Pair with REF-14; one shared capture may satisfy both rows. Hand contact needs cleanup.
☐	☐	☐	RES-06	Loser stands during declaration		v1.0	
☐	☐	☐	RES-07	Winner bows to crowd		v1.0	
☐	☐	☐	RES-08	Winner kneels / arms raised (big win)		v1.0	Finals
☐	☐	☐	RES-09	Loser applauds opponent		v1.0	
☐	☐	☐	RES-10	Opponents embrace after match	Two performers	v1.0	
☐	☐	☐	RES-11	Coach celebrates with fighter	Two roles; embrace or shoulder pat	v1.0	Lift only if separately planned and safely performed; body contact may obscure markers.
☐	☐	☐	RES-12	Podium: step up, receive medal, bow head	Gold, silver, bronze positions	v1.0	Career events
☐	☐	☐	RES-13	Podium photo pose	3 variations	FUTURE	

13 FATIGUE AND STAMINA

Layer-ready breathing and posture changes for low stamina. Keep the lower body consistent with the guard idles.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	FAT-01	Heavy breathing in guard (additive)	Left Lead & Right Lead	v1.0	Upper body only
☐	☐	☐	FAT-02	Hands on hips / knees during stoppage		v1.0	
☐	☐	☐	FAT-03	Slow recovery after big spinning attack	Spin from Left Lead & Right Lead	v1.0	
☐	☐	☐	FAT-04	Wipe face / shake out legs		v1.0	
☐	☐	☐	FAT-05	Stretching leg during stoppage	Left and right	v1.0	

14 REFEREE AND OFFICIALS

Separate suited performer; final character wears referee uniform. Verify command gestures with the current official reference and a qualified referee before final capture. This list names coverage; it does not define the exact signal choreography. Finger counts require separate finger animation.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	REF-01	Referee ready stance between fighters	Loop	v1.0	
☐	☐	☐	REF-02	Charyeot command gesture		v1.0	
☐	☐	☐	REF-03	Kyeongrye command gesture		v1.0	
☐	☐	☐	REF-04	Joon-bi command gesture		v1.0	
☐	☐	☐	REF-05	Shi-jak command gesture (start)		v1.0	
☐	☐	☐	REF-06	Kal-yeo command (step in, separate fighters)	Two fighters optional	v1.0	
☐	☐	☐	REF-07	Kye-sok command gesture (continue)		v1.0	
☐	☐	☐	REF-08	Keu-man command gesture (stop)		v1.0	
☐	☐	☐	REF-09	Gam-jeom declaration (point at fighter, signal)	Toward Chung and toward Hong	v1.0	
☐	☐	☐	REF-10	Knockdown count (counting with fingers)		v1.0	Body capture records arm/hand motion; finger counts must be hand-keyed or captured separately.
☐	☐	☐	REF-11	Time-out / injury stoppage gesture		v1.0	
☐	☐	☐	REF-12	Call medic		v1.0	
☐	☐	☐	REF-13	Video review acknowledgment		FUTURE	
☐	☐	☐	REF-14	Declare winner (raise fighter's hand)	Chung winner and Hong winner	v1.0	Paired with RES shots

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	REF-15	Referee idle walking the mat (follows fighters)	Loop; left and right	v1.0	
☐	☐	☐	REF-16	Judges seated with scoring devices	Loop	v1.0	Judges' tables around the mat (GDD 6.3)

15 COACH AND CORNER

Separate performer in optical marker suit; final character wears coach attire. Use visibility-tested chair, towel and water props or consistent stand-ins.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	COA-01	Coach seated idle (loop)		v1.0	
☐	☐	☐	COA-02	Coach shouting instructions (standing)		v1.0	
☐	☐	☐	COA-03	Coach clapping		v1.0	
☐	☐	☐	COA-04	Coach frustration (hands on head)		v1.0	
☐	☐	☐	COA-05	Coach celebration		v1.0	
☐	☐	☐	COA-06	Coach giving advice between rounds (leaning in)	Paired with CER seated rest	v1.0	
☐	☐	☐	COA-07	Coach handing water / towel		v1.0	

16 MENUS AND CAREER (OUT OF GEAR)

Final fighter wears dobok and belt without sparring gear. Source performer stays in the optical marker suit. Use props/stand-ins for gear-up actions, then align fingers, straps and clothing contacts in cleanup.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	MNU-01	Menu idle: relaxed stand in dobok	3 variations; loop	v1.0	Main menu and fighter profile
☐	☐	☐	MNU-02	Menu idle: arms crossed	Loop	v1.0	
☐	☐	☐	MNU-03	Menu idle: light bounce, shaking out	Loop	v1.0	
☐	☐	☐	MNU-04	Turn to camera and bow		v1.0	Fighter selected
☐	☐	☐	MNU-05	Fighter profile pose: ready stance		v1.0	
☐	☐	☐	MNU-06	Fighter profile pose: guard	Left Lead & Right Lead	v1.0	
☐	☐	☐	MNU-07	Tie belt		v1.0	Share CER-14 source; fingers, knot and belt deformation require additional animation.
☐	☐	☐	MNU-08	Put on and fasten chest protector		v1.0	Gear-up sequence
☐	☐	☐	MNU-09	Put on and fasten helmet		v1.0	
☐	☐	☐	MNU-10	Put on gloves and foot guards		v1.0	
☐	☐	☐	MNU-11	Take off helmet after match		v1.0	
☐	☐	☐	MNU-12	Sit on locker-room bench (loop)		v1.0	
☐	☐	☐	MNU-13	Walk through dojang / locker room	Loop	v1.0	
☐	☐	☐	MNU-14	Grading exam: kneel, receive new belt, bow		v1.0	Rank promotion
☐	☐	☐	MNU-15	Grading exam: examiner presents belt	Examiner performer	v1.0	

17 TRAINING DRILLS

Suited kicker and pad holder. Capture together only after a multi-subject visibility test; otherwise record each role against the same marked target and timing. Record prop dimensions/locations for engine contact alignment.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	TRG-01	Pad holder: hold paddle at body height	Left and right side; loop	v1.0	Partner for kicking drills
☐	☐	☐	TRG-02	Pad holder: hold paddle at head height	Left and right side; loop	v1.0	
☐	☐	☐	TRG-03	Pad holder: absorb kick reaction	Body and head height	v1.0	
☐	☐	☐	TRG-04	Pad holder: moving paddle (step back, present)	Loop	v1.0	
☐	☐	☐	TRG-05	Kicking paddle drill: rapid round kicks	Left Lead & Right Lead	v1.0	
☐	☐	☐	TRG-06	Heavy bag / standing target kicks	Round, side and spin side kick	v1.0	
☐	☐	☐	TRG-07	Reaction target drill (turn and kick)	Left Lead & Right Lead	v1.0	
☐	☐	☐	TRG-08	Shadow sparring loop	Left Lead & Right Lead	v1.0	30 s continuous
☐	☐	☐	TRG-09	Chamber hold balance drill	Lead leg & Rear leg; Left Lead & Right Lead	v1.0	
☐	☐	☐	TRG-10	Stretching: leg swings (front, side)	Left and right	v1.0	
☐	☐	☐	TRG-11	Stretching: splits and hurdler stretch		v1.0	
☐	☐	☐	TRG-12	Jump rope loop		v1.0	
☐	☐	☐	TRG-13	Conditioning: fast feet / ladder footwork	Loop	v1.0	
☐	☐	☐	TRG-14	Instructor demonstration pose (pointing, explaining)		v1.0	Lessons

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	TRG-15	Training partner wipes mirror	One hand then the other; looping circles; enter and exit	v1.0	Wax On Wax Off training easter egg (GDD 6.5). Use a non-reflective stand-in surface and cloth prop; hand contact needs cleanup.

18 CROWDS AND FUTURE SPARRING FORMATS

Venue crowd coverage (FUT-05) is v1.0. Capture the other rows only when their Future format is scheduled. Para sets require appropriate athletes and format-specific rules; youth sets require suitable performers and production arrangements.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	FUT-01	Team Championship: tag in / tag out at the line	Entering and leaving fighter	FUTURE	Team events
☐	☐	☐	FUT-02	Team bench idle and reactions	Loop	FUTURE	
☐	☐	☐	FUT-03	Para taekwondo sparring set (guards, footwork, kicks, reactions)	Per classification; captured with para athletes	FUTURE	Follow WT Para rules (e.g. no head kicks)
☐	☐	☐	FUT-04	Cadet and junior performer set (core idles, kicks, reactions)	Smaller performers	FUTURE	Proportion-accurate youth fighters
☐	☐	☐	FUT-05	Crowd: seated idle, cheer, applaud, groan	Seated idle, cheer, applaud, groan and raise sign; loops	v1.0	v1.0 venue crowds, including score-triggered crowd signs (GDD 6.2 and 6.5). Hand-key fingers and sign contacts.
☐	☐	☐	FUT-06	Commentator desk idle	Loop	FUTURE	Commentary (GDD 13)

19 SLIDING STRIKE VARIATIONS

Record sliding kicks and punches as deliberate variations of the mapped techniques (GDD 7.4). The strike, leg/hand choice, target and points stay the same; the fighter travels through the technique. Capture an actual moving performance with its root path, support-foot pivot and landing.

Directions use the fighter facing the opponent at the start of the take: F forward, B back, L left, R right, FL forward-left, FR forward-right, BL back-left and BR back-right. Use all eight directions for full v1.0 coverage. For turns, keep the starting frame of reference and log the resulting path; slide direction and spin direction are separate variants.

M1 priority: forward/back body side and roundhouse from both stances, with the applicable lead/rear leg choices. Capture their lateral and diagonal variants for v1.0. Add cut, spinning and punch slide coverage for M2. Other rows follow their listed phase. Each row covers multiple variants, not one recording.

For each required direction, log source/destination stance, striking leg or hand, target height, slide displacement, duration, contact frame and take ID. Keep a stationary baseline. Capture the strike during the travel, not only after an entry step. Check stationary-to-slide, slide-to-still, slide-to-guard and fake-cancel transitions. Hit/miss/block and landing coverage can be shared by family; do not multiply every outcome into every combination.

SLD-09 to SLD-13 are derived by default from moving grounded sources plus TRN-12. Accept the assembled jump only after checking its root path, lift-off, landing and facing. Capture a dedicated moving jump if reuse fails. The travelling Tornado Kick needs a unique performance. Never force an awkward direction into a false foot slide; record the natural step/hop solution or flag that variant for a pickup.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	SLD-01	Sliding cut kick	8 directions; lead leg; both stances; body	M2	KCK-08. Record the jam during travel; preserve chamber, support-foot motion and recovery.
☐	☐	☐	SLD-02	Sliding side kick	8 directions; lead/rear; both stances; body/head	M1	KCK-09. M1 forward/back body first; remaining directions/targets by v1.0. Preserve hip turn and landing.
☐	☐	☐	SLD-03	Sliding roundhouse kick	8 directions; lead/rear; both stances; body/head	M1	KCK-01/03/04. M1 forward/back body first. Reuse KCK-05/06 where their direction/technique match; log the source.
☐	☐	☐	SLD-04	Sliding hook kick	8 directions; lead/rear; both stances; body/head	v1.0	KCK-13. Preserve the hooking path, foot pivot and rechamber while travelling.
☐	☐	☐	SLD-05	Sliding axe kick	8 directions; lead/rear; both stances; head	v1.0	KCK-12. Keep the rising/dropping leg path and a controlled travelling landing.
☐	☐	☐	SLD-06	Sliding spin side kick	8 directions; from both stances; body/head	M2	KCK-14/15. Record root travel separately from the turn; check pivot, facing and balance.

RECORDED	CLEANED	IN ENGINE	ID	ANIMATION	VARIANTS TO RECORD	PHASE	NOTES
☐	☐	☐	SLD-07	Sliding spin hook kick	8 directions; from both stances; body/head	M2	KCK-16. Check optical tracking through the turn and the moving support-foot contacts.
☐	☐	☐	SLD-08	Travelling Tornado Kick	8 directions; from both stances; body/head	v1.0	KCK-23. Unique moving capture; log takeoff/landing, root path and facing. Use natural steps/hops.
N/A	☐	☐	SLD-09	Travelling jump side kick	8 directions; lead/rear; both stances; body/head	v1.0	DERIVED: SLD-02 + TRN-12; paired with KCK-32. Verify travel and landing; optional pickup if needed.
N/A	☐	☐	SLD-10	Travelling jump roundhouse	8 directions; lead/rear; both stances; body/head	v1.0	DERIVED: SLD-03 + TRN-12; paired with KCK-26. Verify travel and landing; optional pickup if needed.
N/A	☐	☐	SLD-11	Travelling jump hook kick	8 directions; lead/rear; both stances; body/head	v1.0	DERIVED: SLD-04 + TRN-12; paired with KCK-33. Verify travel and landing; optional pickup if needed.
N/A	☐	☐	SLD-12	Travelling jump axe kick	8 directions; lead/rear; both stances; head	v1.0	DERIVED: SLD-05 + TRN-12; paired with KCK-34. Verify travel and landing; optional pickup if needed.
N/A	☐	☐	SLD-13	Travelling jump spin hook	8 directions; from both stances; body/head	v1.0	DERIVED: SLD-07 + TRN-12; paired with KCK-25. Verify travel and landing; optional pickup if needed.
☐	☐	☐	SLD-14	Sliding straight punch	8 directions; lead/rear hand; both stances; trunk	M2	PNF-01/02/03. X, trunk only. Capture travel through the punch and recovery; reuse PNF-03 for matching backward motion.

CAPTURE SESSION LOG

One row per capture block or calibration change. Link the raw trial folder, subject/template and detailed take manifest. Keep selected takes, rejected/pickup notes, marker changes, FPS, axes/units, frame ranges, video sync and exports together.

DATE	LOCATION	PERFORMERS	SHOT IDS RECORDED	RAW TRIAL FOLDER	NOTES

CAPTURE REFERENCES

Project references: Docs/GDD_Open_Stance.docx v0.11, sections 6.4, 6.5, 7.4, 7.10, 8.6-8.10, 9, 16.2-16.4, 19 and 20.1; Docs/Mocap_Marker_Set.md; Docs/Vicon_Nexus_Test.md; Resources/Models/Animation/README.md.
Vicon subject calibration: https://www.vicon.com/support/faqs/why-should-i-complete-a-subject-calibration-within-nexus/
Vicon processed C3D export: https://vicon-help.atlassian.net/wiki/spaces/Nexus220/pages/897714541/Configure+file+export+pipeline+operations
Before final capture, confirm the installed lab software, marker template, solver, partner tracking and export path with the operator. Approve unresolved guard poses and referee signals with the practitioner/referee. These choices must be settled before their production rows can be signed off.